

9. PARKING – VEHICLE ENTRY & EXIT

FEATURE SET

Vehicle details

Parking slot

Entry time

Exit time

Parking fee

1. PROMPT FOR ALGORITHM

Create a simple and easy-to-understand algorithm for the following college first-year assignment.

Problem Statement:

Parking – Vehicle Entry & Exit

Description:

Design a Vehicle Parking Management System to manage vehicle details, parking slot allocation, entry time, exit time, and parking fee. The system should allow parking staff to record vehicle details, assign a parking slot, record entry and exit times, and calculate the parking fee.

Feature Set:

Vehicle details

Parking slot

Entry time

Exit time

Parking fee

Requirements:

Write the algorithm in numbered step-by-step format.
Start with entering vehicle details.

Show how the parking slot is assigned.
Record the vehicle entry time.
Record the vehicle exit time.
Calculate the parking fee based on the parking duration.
Display the parking fee.
End the process after displaying the parking fee.
Use simple language suitable for first-year students.
Do not add features outside the given Feature Set.

Generate the final algorithm in a clean, well-structured and easy-to-read format.

2. PROMPT FOR FLOWCHART

Create a professional, attractive and presentation-ready FLOWCHART IMAGE for a college first-year assignment.

Problem Statement:

Parking – Vehicle Entry & Exit

Feature Set:

Vehicle details
Parking slot
Entry time
Exit time
Parking fee

Create a complete logical flowchart showing the process from Start to End.

The flow should include:

START

↓

Enter Vehicle Details

↓

Assign Parking Slot

↓
Record Entry Time
↓
Record Exit Time
↓
Calculate Parking Fee
↓
Display Parking Fee
↓
END

Use standard flowchart notation:

Oval → Start / End
Parallelogram → Input / Output
Rectangle → Process
Diamond → Decision, only where logically required
Arrows → Direction of flow

Visual Requirements:

Use proper standard flowchart shapes.
Use clean and straight connecting arrows.
Maintain proper spacing and alignment.
Make all text large, sharp and readable.
Use a modern academic and professional design.
Use subtle attractive colors while maintaining clear flowchart notation.
Avoid overlapping shapes and lines.
Keep the diagram simple and suitable for first-year students.
Do not add unnecessary features.

Add the title:

"Flowchart – Parking – Vehicle Entry & Exit"

Generate ONLY the final high-resolution flowchart image.

3. PROMPT FOR ER DIAGRAM

Create a professional, attractive and presentation-ready ER DIAGRAM as a high-resolution image for a college first-year assignment.

Problem Statement:

Parking – Vehicle Entry & Exit

Feature Set:

Vehicle details
Parking slot
Entry time
Exit time
Parking fee

IMPORTANT:

Use CLASSICAL / CHEN ER DIAGRAM NOTATION.

Use these shapes strictly:

RECTANGLE → Entity

OVAL / ELLIPSE → Attribute

DIAMOND → Relationship

UNDERLINED ATTRIBUTE → Primary Key (PK)

Clearly indicate Foreign Keys (FK)

Straight connector lines between entities, attributes and relationships

Show proper cardinality near relationship lines

Entities and Attributes:

VEHICLE

vehicle_id (PK)
vehicle_number
vehicle_type
owner_name
mobile

PARKING_SLOT

slot_id (PK)
slot_number
slot_type
slot_status

PARKING

parking_id (PK)
vehicle_id (FK)
slot_id (FK)
entry_time
exit_time
parking_fee

Relationships:

VEHICLE — PARKS — PARKING
PARKING_SLOT — ASSIGNED_TO — PARKING

Visual Requirements:

Use rectangular boxes for entities.
Use oval shapes for attributes.
Use diamond shapes for relationships.
Connect every attribute to its corresponding entity.
Connect entities through relationship diamonds.
Clearly show Primary Keys and Foreign Keys.
Show appropriate cardinality such as 1:1 or 1:M.
Keep the diagram balanced and well organized.
Avoid crossing connector lines wherever possible.
Use a clean white background.
Use sharp, readable text.
Use subtle professional colors.
Do not use table-style database diagrams.
Do not use flowchart symbols.
Do not add unnecessary entities or attributes.
Do not use icons or 3D effects.
No watermark.

Add the title:

“ER Diagram – Parking – Vehicle Entry & Exit”

Generate ONLY the final high-resolution ER diagram image.